that explain your observations, because you are as justified as the next guy in claiming that you are at rest and it's everything else that is moving. Einstein, as we have seen, extended this symmetry in a thoroughly unanticipated way by including the speed of light among the observations that would be unaffected by either your motion or the motion of the light's source. This was a stunning move because we ordinarily throw the particulars of an object's speed into the environmental details bin, recognizing that the speed observed generally depends upon the motion of the observer. But Einstein, seeing light's symmetry stream through the cracks in nature's Newtonian façade, elevated light's speed to an inviolable law of nature, declaring it to be as unaffected by motion as the cue ball is unaffected by rotations.

General relativity, Einstein's next major discovery, fits squarely within this march toward theories with ever greater symmetry. Just as you can think of special relativity as establishing symmetry among all observers moving relative to one another with constant velocity, you can think of general relativity as going one step farther and establishing symmetry among all accelerated vantage points as well. This is extraordinary because, as we've emphasized, although you can't feel constant velocity motion, you *can* feel accelerated motion. So it would seem that the laws of physics describing your observations must surely be different when you are accelerating, to account for the additional force you feel. Such *is* the case with Newton's approach; his laws, the ones that appear in all first-year physics textbooks, must be modified if utilized by an accelerating observer. But through the principle of equivalence, discussed in Chapter 3, Einstein realized that the force you feel from accelerating is indistinguishable from the force you feel in a gravitational field of suitable strength (the greater the acceleration, the greater the gravitational field). Thus, according to Einstein's more refined perspective, the laws of physics do *not* change when you accelerate, as long as you include an appropriate gravitational field in your description of the environment. General relativity treats all observers, even those moving at arbitrary non-constant velocities, equally—they are completely symmetric—since each can claim to be at rest by attributing the different forces felt to the effect of different gravitational fields. The differences in the observations between one accelerating observer and another are therefore no more surprising and provide no greater evidence